Zero-Shot Multidegraded Hyperspectral Image Restoration via Intrinsic Image Decomposition

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***Abstract*—Hyperspectral images (HSIs) are often simultaneously degraded by spatial downsampling, complex noise contamination, and pixel loss during acquisition, resulting in spatial blurring and spectral distortion that severely hinder quantitative analysis and downstream applications. Most existing HSI restoration methods are designed for single degradation scenarios and heavily rely on paired supervision, which limits their applicability in unseen extreme environments where ground truth is unavailable. To address these challenges, this article proposes an unsupervised zero-shot HSI restoration (ZSHR) method based on intrinsic image decomposition (IID), grounded in physical imaging principles. Specifically, an HSI is modeled as the multiplicative interaction between reflectance and shading components, and the restoration under multiple degradations is reformulated as a joint estimation problem of intrinsic components under physical consistency constraints. This physics-driven decoupling allows the model to isolate inherent material properties from environmental illumination factors, ensuring high spectral fidelity in extreme scenarios. Accordingly, an end-to-end optimization framework is developed to jointly perform denoising and super-resolution (SR) reconstruction. By incorporating intrinsic consistency constraints between reflectance and shading, frequency-guided feature modulation, and panchromatic-guided cross-modal spatial fidelity constraints, the proposed method achieves high-quality HSI restoration without any external supervision. Extensive experiments on multiple public remote sensing datasets and challenging lunar in situ hyperspectral data demonstrate that the proposed method consistently outperforms existing approaches in terms of spectral fidelity, spatial detail preservation, and downstream classification and unmixing performance.**

***Index Terms*—Hyperspectral image restoration, intrinsic image decomposition (IID), panchromatic-guided restoration, zero-shot restoration (ZSR).**

I. Introduction

HYPERSPECTRAL imagery (HSI) demonstrates immense application value in fields such as remote sensing monitoring [1], [2], deep-space exploration [3], [4], and medical detection [5], [6], owing to its rich spectral and spatial information. However, during the actual acquisition, due to inherent sensor limitations and complex imaging environments, the obtained HSIs are typically subject to intertwined multidegradation issues, including low spatial resolution, noise interference, and pixel loss. Multidegradation phenomena exhibit high coupling in physical mechanisms, leading to simultaneous distortion of spatial structures and spectral reflectance, which severely constrains subsequent high-precision analysis and quantitative applications. Therefore, reconstructing HSIs with both spatial and spectral fidelity from multidegraded observations has become a critical challenge in the field of HSI analysis.

Existing HSI restoration methods can be broadly divided into model-driven optimization algorithms and data-driven deep learning algorithms. Model-driven algorithms typically formulate the restoration task as an ill-posed inverse problem and impose hand-crafted priors-such as global priors based on tensor low-rank [7], [8], [9] and sparsity [10], [11], [12], local smoothness priors [13], [14], [15], [16], and nonlocal self-similarity priors [17], [18], [19] to constrain the solution space. While effective under simplified assumptions, these methods often fail to characterize the complex physical entanglement between spatial illumination and spectral reflection, leading to a loss of inherent spectral characteristics when multiple degradation factors are coupled. In contrast, data-driven deep learning methods directly learn complex mapping relationships from paired training data, typically achieving superior restoration performance [20], [21], [22]. However, their reliance on large-scale, high-quality supervised data and static degradation models severely limits their application in real-world and remote sensing scenarios where paired data is scarce, and degradation conditions vary significantly.

To address these limitations, recent research has gravitated toward the zero-shot restoration (ZSR) paradigm [23], [24], [25], which leverages internal image statistics without requiring external training. Despite their flexibility, most existing ZSR methods remain confined to holistic statistical reconstruction. By treating the HSI as a monolithic data cube under uniform constraints, they fail to distinguish between intrinsic spectral signatures and extrinsic illumination-induced structures. This lack of component-level discrimination inevitably leads to a fundamental tradeoff: oversmoothing spectral signatures or losing fine-grained spatial details, especially under complex multidegradation where noise and blurring are deeply intertwined.

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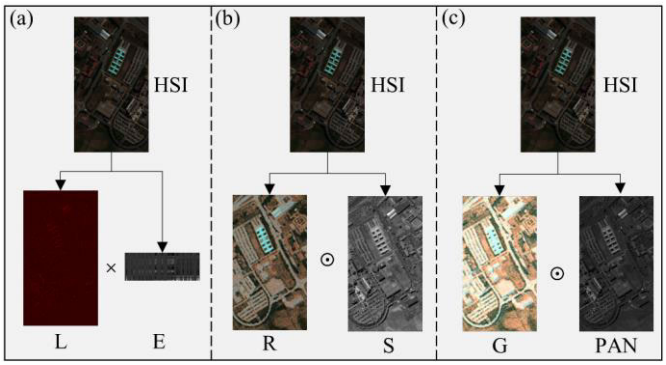


Fig. 1. Modeling paradigms for HSI restoration. (a) Conventional statistical low-rank decomposition of HSI into a low-rank component L and a recovery matrix E. (b) Physics-driven IID, modeling HSI as the multiplicative interaction of reflectance R and shading S. (c) PAN-guided spatial–spectral alignment, where the coefficient matrix G adaptively links PAN and HSI.

This technical limitation becomes particularly critical in hyperspectral remote sensing, especially deep-space in situ exploration, where spectral reflectance is the primary quantity for quantitative analysis. In such environments, the absence of atmospheric scattering creates extreme light-shadow contrasts, further coupling shading effects with inherent material properties. By relying solely on statistical consistency, traditional ZSR methods struggle to decouple environmental illumination from extraterrestrial material spectra. Consequently, while visual quality may be enhanced, the results often suffer from spectral shifts and physical inexplicability, ultimately undermining the reliability of reconstructed data for downstream scientific missions.

To address these challenges, this article proposes a multidegraded HSI ZSR framework predicated on the synergy of panchromatic (PAN)-fusion and intrinsic image decomposition (IID). The cornerstone of our approach is the reformulation of the HSI observation as $I = R \odot S$, where R and S represent the reflectance and shading components, respectively. By shifting the reconstruction from a purely statistical domain to a physically interpretable space, we decouple inherent material properties from spatial illumination artifacts. Fig. 1 illustrates the fundamental differences among three representative modeling paradigms for HSI restoration. Conventional statistical low-rank decomposition [see Fig. 1(a)] treats all degradation effects as entangled residuals, resulting in physically unidentifiable factors. In contrast, IID [see Fig. 1(b)] explicitly separates R and S, providing a physically interpretable decomposition that decouples spectral properties from spatial variations. Incorporating high-resolution PAN [see Fig. 1(c)] further constrains spatial modulation, enhancing structural details while maintaining spectral fidelity. This figure highlights the paradigm shift from purely statistical reconstruction toward physics-driven restoration, which forms the conceptual basis of the proposed method.

To ensure the stability and physical rationality of IID under multidegradation conditions, we introduce a set of intrinsic-guided modeling components that all operate to guide the decoupling in the physical domain. Among these, the intrinsic-guided denoising module (IGDM) and intrinsic-guided super-resolution module (IGSM) are used to stabilize the estimation of reflectance and shading components during denoising and super-resolution (SR) processes, while the coefficient-modulated spatial fidelity model (CMSFM) constrains the PAN-guided spatial enhancement process to avoid spectral distortion caused by cross-modal inconsistencies. The above designs collectively serve the physical assumptions of IID, enabling robust multidegraded HSI restoration without any external training data. Extensive experiments demonstrate that the proposed method achieves superior spatial detail recovery, spectral fidelity, and unprecedented robustness in extreme deep-space scenarios.

The main contributions can be summarized as follows.

1) A zero-shot HSI restoration (ZSHR) paradigm is developed by integrating IID with physical constraints. It leverages spatial–spectral decoupling to maintain intrinsic reflectance purity, providing a reliable reconstruction tool for complex, unlabeled scenarios.
2) An intrinsic-guided modeling framework is developed, integrating specialized architectural modules with joint physical constraints. Specifically, IGDM and IGSM leverage frequency-domain complementarity to isolate noise and inject high-frequency details, while CMSFM is proposed to adaptively align spectral responses with panchromatic guidance. Synergized with intrinsic consistency and structural fidelity losses, this framework ensures a physically consistent restoration.
3) Extensive experiments on multiple remote sensing benchmark datasets and lunar in situ HSI are conducted, demonstrating that the proposed method consistently achieves high spectral fidelity and spatial reconstruction quality under various multidegradation conditions, and exhibits robust generalization performance in downstream tasks.

The remainder of this article is organized as follows. Section II reviews the related work on ZSHR and IID. Section III elaborates on the proposed IID-guided joint restoration framework. Section IV describes the qualitative and quantitative analysis results. Finally, Section V summarizes the work and discusses future research directions.

II. Related Works

A. Zero-Shot HSI Restoration

ZSHR aims to recover degraded HSIs by implicitly or explicitly encoding image priors during a single-observation optimization process, thereby bypassing the need for external paired training data. Existing ZSHR methods can be broadly categorized into implicit prior-based and explicit constraint-based approaches.

Implicit prior-based methods utilize the structural bias of neural networks as a form of regularization. A representative work is the deep image prior (DIP) [26], [27], [28], which demonstrated that an untrained convolutional network can capture underlying statistical information of images through per-image optimization. Niresi and Chi [29] extended the DIP concept to HSI restoration tasks, showcasing its potential for single-image denoising without precollected training datasets.

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To further enhance regularization, recent architectures have introduced dual-branch structures [30] or pyramid backbones [31] to generate pseudo-references and improve convergence. Although implicit prior-based methods offer flexibility, they typically lack explicit physical constraints and external references. Consequently, they tend to treat noise, spatial structural variations, and spectral distortions as statistically indistinguishable perturbations, often struggling to handle the diverse noise types common in HSIs and complex degradation patterns [32], particularly in multidegradation scenarios.

Explicit constraint-based methods address these limitations by embedding physical models or inherent data properties into the optimization objective. A common strategy involves exploiting the low-rank of HSI data [25], [33], [34] through matrix or tensor regularization to enforce spectral consistency. Moving beyond purely statistical priors, some researchers incorporate imaging models or task-specific constraints. For instance, in pansharpening, sensor spectral responses are estimated to enforce imaging-model-based losses [35]. Recently, zero-shot optimization has been coupled with diffusion null-space models [36] or linear spectral unmixing frameworks to provide more rigorous physical grounding. However, while these methods improve fidelity, they rarely consider the intrinsic decomposition of the scene, often failing to disentangle extrinsic illumination effects from intrinsic material reflectance.

B. HSI Intrinsic Image Decomposition

IID models the observed image as the elementwise product of a reflectance component and a shading component, i.e., $I = R \odot S$. For common red–green–blue color images, the basic IID model assumes that the shading component S does not vary with wavelength. However, for HSI, this assumption does not hold, as the formation mechanism of HSI is influenced by sunlight illumination and wavelength-dependent atmospheric interactions (e.g., Rayleigh scattering). Consequently, illumination and shading effects in HSI vary across spectral bands. To address this issue, the hyperspectral IID (Hyper-IID) model explicitly incorporates wavelength dependence into the decomposition process, representing observations as $I = R(\lambda) \odot S(\lambda)$ [37], [38]. This representation provides a physically interpretable decomposition by separating the HSI into two intrinsic components with distinct physical meanings.

In existing studies, the reflectance component is typically regarded as the carrier of material-specific spectral characteristics and is closely related to downstream quantitative analysis, while the illumination or shading component captures spatial structures and geometric variations, which can be used to guide spatial enhancement and resolution recovery [39], [40], [41], [42]. By separating these intrinsic components, Hyper-IID offers a principled approach to decoupling spectral properties from spatial effects, thereby transforming HSI restoration into the estimation and constraint of physically meaningful variables. This characteristic makes IID particularly attractive in unsupervised and physics-guided learning. However, existing HyperIID-based methods are primarily studied under supervised learning or simplified single-degradation assumptions. Their potential for ZSR under complex mixed degradations has not been fully explored, nor have they been systematically integrated with PAN-guided spatial enhancement within a unified physical modeling framework.

III. Proposed Method

A. Problem Formulation

During acquisition, HSIs are frequently subjected to multiple intertwined degradation phenomena due to limitations in imaging conditions and sensor specifications. These effects, including spatial downsampling, noise contamination, and pixel-level missing observations can be jointly formulated as: $N = \text{Mask}(H_{\downarrow} + \varepsilon)$, where $H \in \mathbb{R}^{h \times w \times b}$ denotes the latent high-quality HSI, where $(\cdot)_{\downarrow}$ denotes the spatial downsampling operator, $N \in \mathbb{R}^{(h/s) \times (w/s) \times b}$ represents the degraded observation under a spatial downsampling factor $s > 1$; ε accounts for stochastic acquisition noise; and Mask$(\cdot)$ models the binary occlusion or invalid pixel measurements.

From the perspective of imaging physics, the captured hyperspectral signal is not solely determined by the intrinsic spectral reflectance of ground objects; rather, it is the result of a synergistic interaction between surface reflectance and spatially varying illumination conditions. Based on the IID principle, an HSI H can be factorized as the elementwise product of a reflectance component and a shading component: $H = R \odot S$, where $R \in \mathbb{R}^{h \times w \times b}$ characterizes the wavelength-dependent reflectance properties, encoding the intrinsic spectral essence of the materials, and $S \in \mathbb{R}^{h \times w}$ depicts the illumination and topography-induced shading effects that dominate the spatial morphology. Under multidegradation scenarios, although noise and spatial blurring inevitably cause spectral distortion and subpixel mixing, a well-defined IID factorization helps shield the reflectance component from dominant illumination-related spatial noise. By isolating these primary spatial interferences, the framework significantly reduces the complexity of restoring spectral purity and intrinsic signatures. Furthermore, the PAN image acts as a structural anchor, providing physically interpretable priors of geometry and illumination to guide shading estimation. This synergy ensures the robust restoration of fine-grained spatial details even under severe degradation.

Consequently, the research objective of this article shifts from the direct estimation of H to a physically grounded decoupling task: the joint inference of intrinsic components R and S in a zero-shot manner. By utilizing the PAN image as a structural anchor for S and enforcing spectral consistency on R, this framework reformulates HSI restoration as the recombination of its intrinsic components.

B. Overall Framework

As illustrated in Fig. 2, the proposed method adopts a two-stage progressive optimization framework, consisting of a pretraining stage and a main training stage. Each stage comprises a sequential pipeline of denoising (DN) and SR. By iteratively executing IID-guided restoration across different spatial scales, the framework progressively stabilizes the decoupling process and refines the estimation of intrinsic components under zero-shot conditions.

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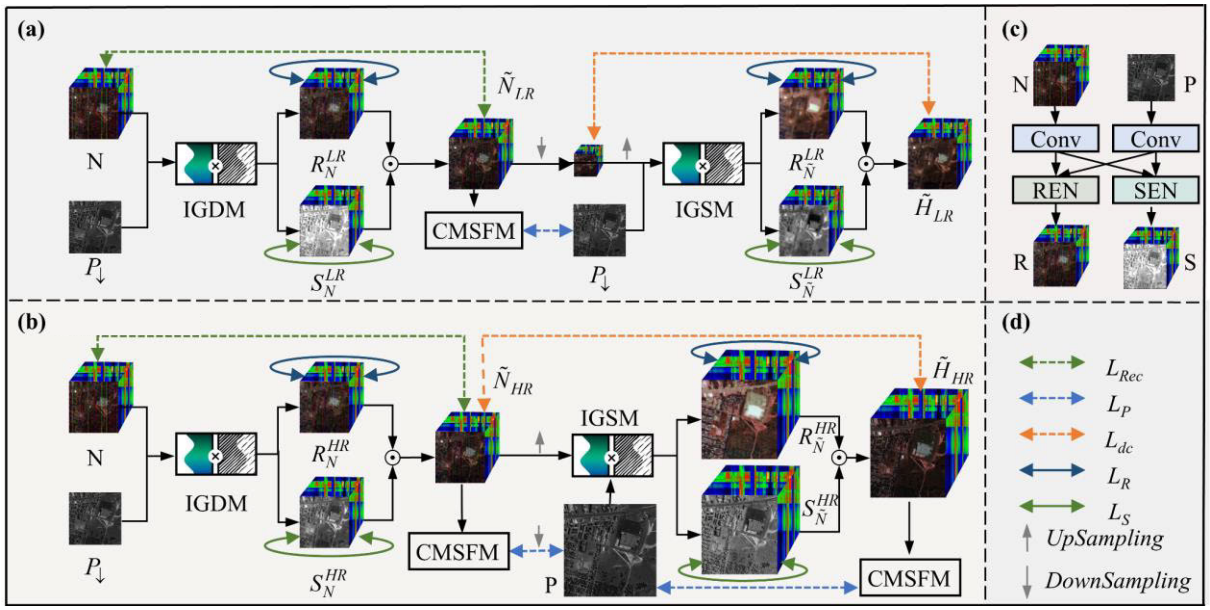


Fig. 2. Overall architecture of the proposed unsupervised ZSHR framework. (a) Pretraining. (b) Main training. (c) IGDM/IGSM. (d) Notations.

1) Pretraining Stage: In this stage, the IGDM takes the degraded HSI N and the downsampled PAN image $P_\downarrow$ (which matches the spatial resolution of N) as inputs to perform intrinsic decomposition under noisy, LR conditions

$$R_N^{LR}, S_N^{LR} = \text{IGDM}\left(N, P_\downarrow\right) \tag{1}$$

where R_N^{LR} and S_N^{LR} represent the estimated reflectance and shading components, respectively. The denoised HSI at the LR scale, denoted as $\tilde{N}_{LR}$, is reconstructed as the Hadamard product of the aforementioned intrinsic components: $\tilde{N}_{LR} = R_N^{LR} \odot S_N^{LR}$.

Subsequently, to simulate the HR reconstruction task within the LR manifold, $\tilde{N}_{LR}$ is further downsampled to $\tilde{N}_{LR\downarrow} \in \mathbb{R}^{(h/s^2)\times(w/s^2)\times b}$. This downsampled HSI, along with $P_\downarrow$, is fed into the IGSM to perform IID-guided reconstruction

$$R_{\tilde{H}}^{LR}, S_{\tilde{H}}^{LR} = \text{IGSM}\left(\left(\tilde{N}_{LR\downarrow}\right)_\uparrow, P_\downarrow\right), \quad \tilde{H}_{LR} = R_{\tilde{H}}^{LR} \odot S_{\tilde{H}}^{LR} \tag{2}$$

where $(\cdot)_\uparrow$ denotes the upsampling operation to align the resolutions of the input HSI and PAN images. The optimized components $R_{\tilde{H}}^{LR}$ and $S_{\tilde{H}}^{LR}$ represent the refined intrinsic estimations at the current scale.

To ensure spatial fidelity, the CMSFM is introduced. This module estimates a spectral response coefficient matrix G to project the restored HSI onto the PAN domain, establishing a physically consistent HSI–PAN constraint

$$G_{\tilde{N}}^{LR} = \text{CMSFM}\left(\tilde{N}_{LR}, P_\downarrow\right). \tag{3}$$

2) Main-Training Stage: The workflow in the main-training stage mirrors the pretraining stage with two critical distinctions. First, the denoised result $\tilde{N}_{HR}$ is fed directly into the IGSM without the intermediate downsampling step. Second, the CMSFM operates concurrently on both the denoised HSI $\tilde{N}_{HR}$ and the final reconstructed HSI $\tilde{H}_{HR}$ to leverage more comprehensive spatial guidance from the original HR PAN image P

$$G_{\tilde{N}}^{HR} = \text{CMSFM}\left(\tilde{N}_{HR}, P_\downarrow\right), G_{\tilde{H}}^{HR} = \text{CMSFM}\left(\tilde{H}_{HR}, P\right). \tag{4}$$

C. Loss Functions

To effectively implement the Hyper-IID physical modeling under zero-shot conditions, this article constructs a set of mutually complementary loss functions centered on the core assumption of reflectance–shading decoupling. These functions regulate the optimization process across three hierarchical levels: 1) image reconstruction losses; 2) intrinsic constraint losses, comprising the reflectance locality loss and shading structure loss; and 3) pan-spectra decoupling fidelity loss. Finally, an overall loss formulation is established to unify these constraints into a progressive optimization strategy, enabling high-fidelity HSI restoration without external supervision.

1) Image Reconstruction Losses: To anchor the intrinsic decomposition within a physically feasible space and maintain scale-invariant consistency, we introduce logarithmic-domain reconstruction and downsampling consistency losses. Following the multiplicative IID model, the IGDM outputs satisfy $\log N \approx \log R_N + \log S_N$. Accordingly, the logarithmic-domain reconstruction loss is defined as

$$\mathcal{L}_{\text{rec}} = \|\log R_N + \log S_N - \log N\|_1. \tag{5}$$

This loss serves as a physical anchor for the IID process, preventing degenerate solutions during zero-shot optimization. Furthermore, to handle the ill-posed nature of unsupervised SR, a downsampling consistency loss is employed to enforce that the super-resolved HSI remains consistent with its LR observation under the degradation model

$$\mathcal{L}_{\text{dc}} = \|\tilde{N} - \tilde{H}_\downarrow\|_F^2. \tag{6}$$

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2) Reflectance Locality Loss: In the Hyper-IID framework, the reflectance component represents the intrinsic spectral signatures of surface materials, which typically exhibit piecewise smoothness: spatial continuity within homogeneous objects and sharp transitions at material boundaries. To characterize this property, we design a reflectance locality loss

$$\mathcal{L}_{\mathrm{R}} = \| \exp\left(-|A\left[\mathcal{N}\left(I\right)\right]|\right) \odot \nabla^2 R\|_F^2 \tag{7}$$

where R denotes the reflectance output from either IGDM or IGSM, and I is the corresponding input HSI. $A[\mathcal{N}(I)]$ computes the spectral angle mapper (SAM) within an eight-neighborhood to evaluate local spectral similarity, while $\nabla^2 R$ captures the second-order gradients of the reflectance. In spectrally similar regions, the weighting term approaches unity, thereby suppressing spatial oscillations. Conversely, at material boundaries where spectral differences are significant, the constraint is adaptively relaxed to preserve sharp edges.

3) Shading Structure Loss: The shading component reflects spatial variations induced by geometric topography and illumination, which should align with brightness gradients but exclude edges caused by abrupt material changes. We formulate the shading structure loss as

$$\begin{aligned}\mathcal{L}_{\mathrm{S}} = \| \exp\left(-|\nabla^2 I|\right) \odot |\nabla^2 S| \\ + |\nabla^2 I| \odot \exp\left(-\left(|A\left[\mathcal{N}\left(I\right)\right]| |\nabla^2 S|\right)\right) \|_F^2\end{aligned} \tag{8}$$

where S signifies the illumination components derived from IGDM and IGSM across both stages. This loss consists of two complementary constraint terms: the first term suppresses spatial variations in the shading component in flat image regions, ensuring continuity of illumination; the second term, in image edge regions, allows discontinuities in the shading component only when the edges do not correspond to material changes. By introducing spectral similarity as a discriminative basis, this loss effectively avoids misinterpretation of material edges by the shading component, thereby promoting stable decoupling of reflectance and shading components.

4) Pan-Spectra Decoupling Fidelity Loss: In the joint processing of HSI and PAN, a physical mapping governed by the sensor's spectral response characterizes their relationship. Specifically, the HSI can be modeled as the spectral modulation of the PAN image. Building upon this imaging mechanism, we formulate the cross-modal structural consistency as a variational optimization problem

$$\min_{\tilde{I},G} \left(\lambda_1 \|\tilde{I} - G \odot \hat{P}\|_F^2 + \lambda_2 \mathcal{R}\left(G\right)\right) \tag{9}$$

where $\tilde{I}$ denotes the HSI output (either $\tilde{N}$ or $\tilde{H}$) at the current stage, and $\mathcal{R}(G)$ denotes the regularization term that characterizes the structural prior of the coefficient matrix G. In our framework, the spectral response coefficient matrix G is dynamically predicted by the CMSFM module rather than being a preset parameter. Inspired by the DIP [26], [28], we treat G as a latent variable parameterized by the network structure itself. By optimizing G within a convolutional framework, the model leverages the inductive bias of CNNs, which inherently favors the reconstruction of structured, low-entropy patterns over stochastic noise. Consequently, the explicit regularization $\mathcal{R}(G)$ can be effectively absorbed into the network architecture, as the architectural constraints provide an implicit regularization that restricts the solution space of G. This leads to the simplified pan-spectra decoupling fidelity loss

$$L_P = \|\tilde{I} - G \odot \hat{P}\|_F^2 \tag{10}$$

where $\hat{P}$ represents the multichannel extension of the PAN image matching the spatial resolution of $\tilde{I}$. This mechanism prevents the optimization from collapsing into a trivial solution (e.g., $G = \tilde{I} \oslash \hat{P}$) and ensures that G captures physically meaningful spatial–spectral correlations.

5) Overall Loss Formulation: The aforementioned loss functions are integrated into a two-stage progressive optimization strategy to ensure robust convergence under zero-shot conditions.

The pretraining stage establishes the intrinsic decomposition baseline within the LR manifold. The total pretraining loss $\mathcal{L}_{\mathrm{pre}}$ is formulated as

$$\mathcal{L}_{\mathrm{pre}} = \mathcal{L}_{\mathrm{IID}}^{LR} + \lambda_P \mathcal{L}_P^{LR} + \lambda_{dc} \mathcal{L}_{dc}^{LR} + \lambda_{rec} \mathcal{L}_{rec}^{LR} \tag{11}$$

where $\mathcal{L}_{\mathrm{IID}}^{LR}$ represents the combined intrinsic constraints for both DN and SR branches

$$\mathcal{L}_{\mathrm{IID}}^{LR} = \sum_{k \in \{N, \tilde{N}\}} \left[\lambda_R \mathcal{L}_R\left(R_k^{LR}, k\right) + \lambda_S \mathcal{L}_S\left(S_k^{LR}, k\right)\right]. \tag{12}$$

In this stage, the PAN fidelity loss $\mathcal{L}_P^{LR}$ and the cross-scale consistency loss $\mathcal{L}_{dc}^{LR}$ are defined as

$$\mathcal{L}_P^{LR} = \mathcal{L}_P\left(\tilde{N}_{LR}, G_{\tilde{N}}^{LR}, P_{\downarrow}\right), \mathcal{L}_{dc}^{LR} = \mathcal{L}_{dc}\left(\tilde{N}_{LR}, \tilde{H}_{LR}\right). \tag{13}$$

Building upon the pretrained weights, the main training stage cascades the intrinsic refinement to the target resolution. The total loss $\mathcal{L}_{\mathrm{main}}$ for this stage is formulated as: $\mathcal{L}_{\mathrm{main}} = \mathcal{L}_{\mathrm{IID}}^{HR} + \lambda_P \mathcal{L}_P^{HR} + \lambda_{dc} \mathcal{L}_{dc}^{HR} + \lambda_{rec} \mathcal{L}_{rec}^{HR}$ In this formulation, the dual PAN fidelity loss is defined as

$$\mathcal{L}_P^{HR} = \mathcal{L}_P\left(\tilde{N}_{HR}, G_{\tilde{N}}^{HR}, P_{\downarrow}\right) + \mathcal{L}_P\left(\tilde{H}_{HR}, G_{\tilde{H}}^{HR}, P\right). \tag{14}$$

The cycle-consistency and reconstruction anchors at this stage are denoted as $\mathcal{L}_{dc}^{HR} = \mathcal{L}_{dc}(\tilde{N}_{HR}, \tilde{H}_{HR})$ and $\mathcal{L}_{rec}^{HR} = \mathcal{L}_{rec}(R_{\tilde{N}}^{HR}, S_{\tilde{N}}^{HR}, N_{HR})$, respectively.

The framework is optimized via a two-stage training strategy to ensure a stable transition from structural initialization to high-resolution refinement. This paradigm is consistent with established coarse-to-fine training schemes in hyperspectral restoration [20], [25].

D. Model Architecture

The proposed architecture is specifically engineered to instantiate the physical paradigm described in Section III-A. Rather than treating HSI restoration as a direct mapping problem, the network is structured as a decoupling-and-fusion pipeline. By isolating the intrinsic reflectance R and shading S, the model ensures that the high-frequency spatial structures from the PAN image can be injected into the HSI without compromising the spectral purity inherent in the low-frequency reflectance component.

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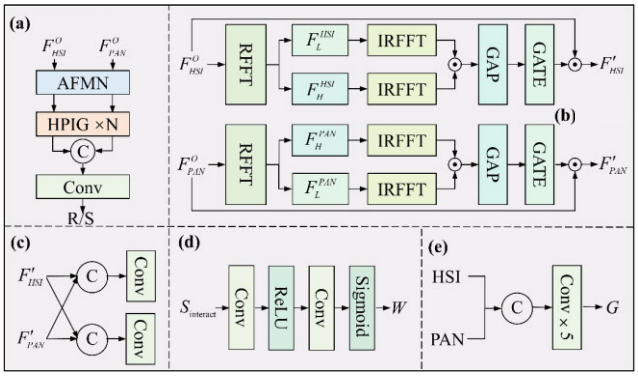


Fig. 3. Detailed structures of (a) REN/SEN, (b) AFMN, (c) HPIG, (d) GATE, and (e) CMSFM.

1) IGDM and IGSM: As illustrated in Fig. 3(a), both the IGDM and the IGSM share a unified architecture comprising a reflectance estimation network (REN) and a shading estimation network (SEN). These components employ a dual-stream multilevel fusion framework, where HSI and PAN features are jointly processed to ensure mutual guidance.

Initially, the input HSI and PAN images are processed by independent convolutional layers to extract initial spectral features F^O_{HSI} and spatial features F^O_{PAN}, respectively. While previous studies have explored frequency-domain processing for HSI denoising [43], [44], destriping [45], and adaptive restoration [46], our asymmetric frequency modulation network (AFMN), as detailed in Fig. 3(b), focuses on leveraging frequency decoupling to manifest the physical properties of intrinsic components. From a physical perspective, the reflectance R is predominantly composed of low-frequency, piecewise smooth variations, whereas the shading S encapsulates high-frequency topographic and textural information.

Specifically, as shown in Fig. 3(b), AFMN utilizes the real-valued fast Fourier transform (RFFT) combined with Gaussian low-pass and high-pass filters to isolate structural features F_L and detail features F_H. This decoupling allows the low-frequency stream to focus on spectral purity, while the high-frequency stream selectively absorbs spatial priors from the PAN image.

Then, as illustrated in Fig. 3(c), the HSI–PAN interactive guidance (HPIG) module utilizes the nonlinear interaction between F_L and F_H to generate a frequency-aware channel attention vector W. By computing the product interaction $S_{\text{interact}} = F_L \odot F_H$, the module encodes the coexistence relationship between structural and textural information. The resulting weight W is calculated via the Gate mechanism shown in Fig. 3(d), $W = \sigma(\text{Gate}(\text{GAP}(S_{\text{interact}})))$. The initial feature streams are then multiplicatively gated as

$$F'_{HSI} = F^O_{HSI} \odot W_{HSI}, F'_{PAN} = F^O_{PAN} \odot W_{PAN}. \quad (15)$$

This mechanism ensures that the fused features prioritize channels that exhibit physically meaningful frequency correlations. After N stacked layers, the features are concatenated and mapped to the final intrinsic components via a tail convolutional sequence. To strictly adhere to IID assumptions, the REN and SEN employ tailored output activations. The REN uses a Sigmoid function to constrain $R \in [0, 1]$, ensuring spectral consistency. For the SEN, a LeakyReLU activation is adopted to capture a wide dynamic range of illumination while preventing gradient vanishing.

2) Coefficient-Modulated Spatial Fidelity Model: The CMSFM is designed to learn the latent spectral response coefficient tensor G, which bridges the gap between the HSI and PAN domains. Each element $G_c(\mathbf{x})$ represents the pixel-level adaptive contribution of the cth HSI band to the PAN intensity at position $\mathbf{x}$. Adopting a lightweight fully convolutional backbone, CMSFM processes the concatenated HSI and PAN inputs to predict $G \in \mathbb{R}^{h\times w\times b}$. Unlike traditional fusion methods that use fixed spectral response functions, CMSFM provides pixel-level adaptive modeling. This design acknowledges that sensor characteristics and local spatial contexts may vary, thereby ensuring that the PAN-derived spatial constraints are both physically consistent and scene-adaptive.

IV. Experiments

A. Dataset

This experiment employs two public HSI datasets, Pavia University (PU) and Urban (UR), as well as a lunar HSI dataset (YT) for performance evaluation. The PU and UR datasets are used to assess the restoration performance of the method in typical urban scenes, while the YT dataset is used to verify the universality and physical consistency of the proposed method under non-Earth imaging conditions and different illumination mechanisms.

1) Pavia University: Acquired over the University of Pavia, Northern Italy, this dataset contains 610 × 340 pixels with 103 spectral bands (430–860 nm) and nine ground object classes.

2) UR: Consists of 307×307 pixels with 210 spectral bands (400–2500 nm); after removing water absorption and low-SNR bands, 162 bands remain. The dataset primarily includes four endmember components.

3) YT: Collected in situ on the lunar surface by the Yutu-2 rover during the Chang'e-4 mission using the Visible and Near-Infrared Imaging Spectrometer (VNIS). VNIS employs acousto-optic tunable filter technology for spectral scanning. The original dataset contains 256 × 256 pixels and 100 bands covering 450–950 nm. Radiance data were converted to reflectance via solar calibration [47].

B. HSI Simulated Degradation Schemes

To evaluate the robustness of HSI DN and SR models, we designed four noise cases simulating spatial resolution degradation, pixel loss, and multisource noise. In all cases, noise is first added according to the specified model, followed by 4× spatial downsampling.

1) Case 1: Independent and identically distributed Gaussian noise. Baseline scenario with independent Gaussian noise (σ = 10%) applied to all bands and pixels.

2) Case 2: Non-independent and identically distributed (Non-IID) + stripe noise: 2/3 of bands receive Gaussian noise (σ = 5%–15%), and 1/3 of bands receive stripe noise with stripe widths randomly 15%–25% of the image width.

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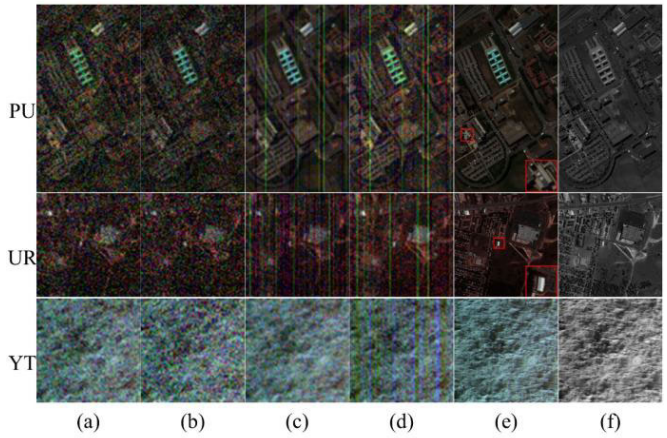


Fig. 4. Visualization of simulated degradation cases and source data. (a)–(d) Simulated HSI under Case 1 to Case 4. (e) Original ground-truth HSI. (f) Corresponding PAN image.

3) Case 3: Medium-intensity multisource noise. It builds on Case 2 by adding deadline noise (random zeroed columns) and impulse noise to 15% of pixels, simulating sensor faults and transmission errors.

4) Case 4: High-intensity multisource noise, same as Case 3, but with higher contamination: impulse noise of 35%, stripe and deadline noise of 35%–45%, testing model robustness under extreme conditions.

We present visualization results of the datasets used in this article and the four degradation cases in Fig. 4.

C. Competing Methods and Quality Assessment

1) Competing Methods: Since most current methods are restoration algorithms designed for single degradation problems, with little research on the multidegradation phenomena mentioned in this article, this article divides the multidegraded HSI restoration task into two steps: DN and SR. Specifically, for the DN stage, four state-of-the-art unsupervised methods are selected as benchmarks: BALMF [9], NGMeet [19], PWRCTV [15], and ATNN [48]. For the subsequent SR stage, we employ two novel diffusion-based frameworks: PLRDiff [49] and HIDiff [23]. Notably, PLRDiff utilizes the PAN image as a spatial prior, whereas HIDiff relies solely on the HSI. Furthermore, we include Hipandas [25], a representative end-to-end HSI restoration model, to evaluate the performance against integrated restoration strategies.

2) Evaluation Metrics: To comprehensively and objectively quantitatively compare the restoration quality of HSIs, we adopt two groups of evaluation metrics, targeting image fidelity and downstream high-level application performance, respectively.

3) Image Fidelity Metrics: We employ four metrics to assess the similarity and fidelity between restored HSIs and ground truth in spatial and spectral domains: peak signal-to-noise ratio (PSNR), structural similarity (SSIM), Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), and SAM. Higher PSNR and SSIM, along with lower ERGAS and SAM, indicate superior image restoration results.

4) Downstream Application Performance Metrics: To further evaluate the support capability of HSI restoration results for preserving ground object information in subsequent tasks, we adopt two groups of metrics for ground object classification and spectral unmixing. For classification results, we use overall accuracy (OA), average accuracy (AA), and Kappa coefficient for quantitative evaluation, where higher OA, AA, and Kappa coefficients represent more accurate classification performance. For spectral unmixing results, we use root-mean-square error (RMSE) and correlation coefficient (CC) for assessment, where lower RMSE and higher CC indicate higher consistency between the unmixed abundance maps and ground truth.

D. Network Training and Experimental Setup

Experiments are conducted on a computer equipped with an RTX 3090 GPU. Training consists of pretrain and main-train stages, comprising 800 and 1000 epochs, respectively, with the Adam optimizer learning rate set to 10^{-4}. λ_R and λ_S are set to 0.7 and 0.5, respectively, while the weights of the remaining loss functions are set to 1.

E. Experimental Results

1) Multidegraded HSI Restoration Results: We compared the proposed method with six other competing methods on three datasets. Tables I–III record the quantitative results on the PU, UR, and YT datasets, respectively. Among them, A, B, N, P, and UH represent ATNN, BALMF, NGMeet, PWRCTV, and Hipandas, while PD and HD represent the PLRDiff and HIDiff methods, respectively. Tables I and II summarize the quantitative performance on the PU and UR datasets across all four multidegradation scenarios. The proposed method consistently achieves optimal or suboptimal results in all metrics. In cases where our method is slightly outperformed, the margin is negligible (less than 1.5%), while it significantly outperforms competing methods in most other indicators. These results underscore the robust generalization and multidegradation mitigation capabilities of our framework when processing standard satellite remote sensing imagery. Table III presents the quantitative evaluation on the YT dataset. Compared to non-end-to-end composite strategies, the proposed method achieves the best performance in nearly all scenarios. Composite baselines exhibit substantial performance degradation in severe scenarios, with PSNR values frequently dropping below 20 dB and SSIM falling below 0.8.

Notably, existing methods often lack consistency across different metrics. For instance, the "HD" baseline performs well in spectral similarity but suffers from severe structural distortion. This discrepancy can be attributed to the lack of high-resolution PAN guidance, essential for anchoring geometric structures under the unique imaging characteristics of the lunar surface, such as extreme shadow contrasts and sparse spectral features. Furthermore, while the UH method occasionally yields slightly higher PSNR and lower ERGAS, it consistently falls behind our method in SAM and SSIM. This indicates that by explicitly modeling the IID process, our method better resolves the intrinsic reflectance properties

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TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS UNDER VARIOUS DEGRADATION CASES ON THE PU DATASET

	Methods	A+PD	B+PD	N+PD	P+PD	A+HD	B+HD	N+HD	P+HD	UH	Ours
Case1	PSNR↑	24.92	24.72	28.35	26.01	26.14	26.29	26.17	26.21	29.98	**30.75**
	SSIM(%)↑	75.51	68.70	81.89	76.77	62.73	63.88	62.18	62.19	83.53	**84.79**
	ERGAS↓	26.21	25.08	20.04	22.03	31.59	31.52	29.94	31.18	20.39	**18.79**
	SAM↓	10.20	10.13	7.67	8.29	7.62	7.78	7.65	7.49	7.30	**6.68**
Case2	PSNR↑	26.40	26.66	27.44	26.30	26.60	26.40	26.22	26.67	30.32	**31.05**
	SSIM(%)↑	81.96	80.01	81.10	80.78	66.43	66.38	64.32	65.87	84.71	**86.54**
	ERGAS↓	22.58	25.76	20.64	19.34	30.04	35.75	29.91	28.62	19.70	**18.32**
	SAM↓	8.78	8.39	8.84	7.15	7.08	8.34	8.09	6.87	6.84	**6.34**
Case3	PSNR↑	26.41	26.71	27.31	25.20	26.55	26.11	26.03	26.68	29.97	**30.89**
	SSIM(%)↑	81.83	78.78	80.32	79.14	65.93	66.19	64.26	65.99	83.82	**86.40**
	ERGAS↓	22.63	25.01	20.80	19.97	30.21	35.07	30.14	29.60	20.39	**18.81**
	SAM↓	8.76	8.31	8.89	7.55	7.15	8.25	8.16	6.98	7.16	**6.51**
Case4	PSNR↑	26.53	26.81	27.61	25.65	26.50	26.51	26.06	26.51	29.41	**30.44**
	SSIM(%)↑	81.74	79.69	81.72	79.69	65.82	66.64	63.39	64.89	82.43	**85.66**
	ERGAS↓	22.79	23.00	22.23	**21.21**	32.08	33.07	32.09	31.42	21.63	21.97
	SAM↓	8.87	8.23	8.70	7.65	7.65	7.97	8.57	7.34	7.63	**7.21**

TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT METHODS UNDER VARIOUS DEGRADATION CASES ON THE UR DATASET

	Methods	A+PD	B+PD	N+PD	P+PD	A+HD	B+HD	N+HD	P+HD	UH	Ours
Case1	PSNR↑	24.92	18.85	24.88	23.65	24.93	24.86	24.94	25.04	28.40	**29.49**
	SSIM(%)↑	75.51	54.79	77.94	74.67	56.73	58.01	55.87	55.80	80.01	**84.37**
	ERGAS↓	26.21	34.18	25.16	26.49	37.26	37.27	35.69	37.37	25.65	**23.19**
	SAM↓	10.20	13.10	9.56	9.47	8.93	8.88	9.35	8.94	8.99	**8.42**
Case2	PSNR↑	26.40	22.56	23.12	25.45	25.40	24.74	25.06	25.32	28.76	**29.69**
	SSIM(%)↑	81.96	72.80	70.01	79.43	60.07	59.22	57.74	58.47	81.20	**85.85**
	ERGAS↓	**22.58**	30.07	29.72	24.01	35.63	40.61	35.45	36.05	24.89	22.97
	SAM↓	8.78	10.01	14.93	8.50	**8.34**	9.00	9.29	8.50	8.42	8.40
Case3	PSNR↑	26.41	21.23	22.94	23.99	25.36	24.14	25.15	25.52	28.55	**29.76**
	SSIM(%)↑	81.83	73.10	70.73	79.01	59.77	59.99	56.31	59.17	80.58	**86.20**
	ERGAS↓	**22.63**	31.88	29.90	24.51	35.79	40.72	35.51	36.11	25.42	**22.99**
	SAM↓	8.76	10.27	15.01	8.27	**8.35**	9.13	9.50	8.51	8.86	8.40
Case4	PSNR↑	26.53	22.37	22.47	25.41	25.36	24.92	24.91	25.25	28.18	**29.17**
	SSIM(%)↑	81.74	71.65	69.22	78.98	59.63	59.71	56.70	57.77	79.39	**84.62**
	ERGAS↓	22.79	28.22	29.31	24.42	35.86	38.22	35.68	36.37	26.40	**24.63**
	SAM↓	8.87	10.25	14.10	8.79	**8.41**	8.76	9.54	8.60	9.40	8.74

TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT METHODS UNDER VARIOUS DEGRADATION CASES ON THE YT DATASET

	Methods	A+PD	B+PD	N+PD	P+PD	A+HD	B+HD	N+HD	P+HD	UH	Ours
Case1	PSNR↑	31.50	28.86	32.21	30.58	26.66	26.34	27.05	16.76	33.12	**33.89**
	SSIM(%)↑	87.79	78.84	88.90	85.47	58.85	59.10	60.70	18.31	90.07	**92.08**
	ERGAS↓	10.07	13.83	9.05	11.82	16.51	16.81	15.91	61.88	8.30	**7.39**
	SAM↓	5.21	6.98	4.25	5.61	3.97	4.39	3.75	11.89	4.18	**3.58**
Case2	PSNR↑	32.18	29.62	28.51	31.28	26.73	18.76	26.62	27.06	**33.50**	32.83
	SSIM(%)↑	87.77	84.78	79.48	89.24	60.91	36.09	61.24	60.04	90.33	**90.74**
	ERGAS↓	10.71	13.86	23.78	10.20	16.53	50.17	16.63	15.95	**8.01**	8.51
	SAM↓	5.51	6.54	10.65	4.81	4.36	10.18	4.63	3.99	4.04	**3.96**
Case3	PSNR↑	31.90	30.12	29.94	31.10	22.29	27.03	26.71	23.79	**33.26**	**32.56**
	SSIM(%)↑	87.17	84.02	81.43	87.75	53.97	62.09	61.36	53.54	90.19	**90.60**
	ERGAS↓	11.21	13.21	18.29	10.38	27.52	15.92	16.56	26.24	**8.26**	9.01
	SAM↓	5.88	6.47	10.02	5.06	10.79	4.26	4.55	4.53	4.19	**4.18**
Case4	PSNR↑	30.63	29.36	28.57	30.95	24.38	19.35	26.50	26.29	30.75	**30.95**
	SSIM(%)↑	87.28	84.38	77.21	85.50	56.03	44.15	60.20	58.41	**89.45**	88.55
	ERGAS↓	14.34	19.33	31.68	51.68	25.99	33.83	18.89	18.03	13.92	13.87
	SAM↓	6.92	7.47	11.71	10.16	6.71	6.11	5.51	5.96	5.66	**5.44**

of the lunar regolith, demonstrating unique adaptability to extreme extraterrestrial environments.

To evaluate spectral fidelity at a finer granularity, Fig. 5 illustrates the bandwise PSNR plots for Case 1 across the

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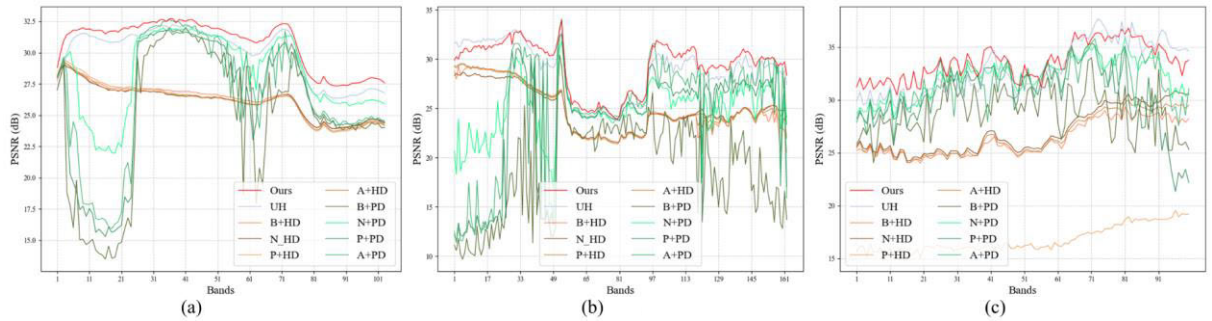


Fig. 5. Bandwise PSNR performance comparison under the Case 1 scenario. (a) PU dataset. (b) UR dataset. (c) YT dataset.

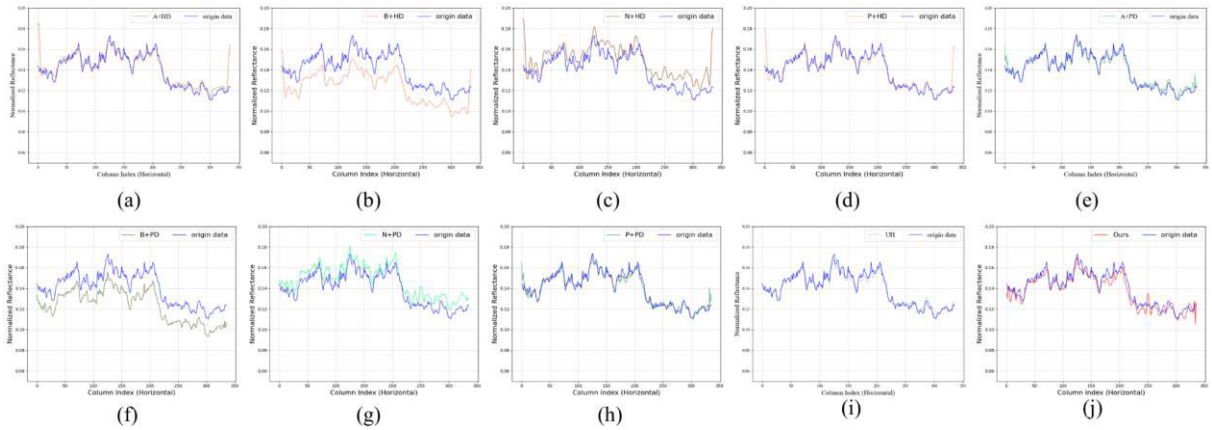


Fig. 6. Vertical profiles of the 36th band for the PU dataset in the mixed noise scenario (Case 2). (a)–(j) A+HD, B+HD, N+HD, P+HD, A+PD, B+PD, N+PD, P+PD, UH, and our proposed method, respectively. The blue line denotes the original high-quality HSI (ground-truth), while other colors represent the results of the corresponding methods.

three datasets. Our method achieves superior PSNR across virtually the entire spectrum, confirming its advantages in cross-band stability. An interesting observation is that the trend of the bandwise curves is highly correlated with the underlying SR framework. Taking the PU dataset [see Fig. 5(a)] as an example, baselines utilizing the PD framework exhibit significantly lower restoration capacity in the low-wavelength regime (Bands 1–20), where PSNR remains below 20 dB. This phenomenon likely stems from the lower signal energy and higher noise sensitivity inherent in short-wavelength bands. Conversely, in the mid-to-high wavelength range (Bands 30–60), their performance improves to levels comparable to our method, indirectly validating the effectiveness of deep network-based spectral processing. Nevertheless, the proposed method maintains a consistent lead, demonstrating that the integration of IID constraints and frequency-aware guidance ensures more stable and high-fidelity restoration across the entire spectral range.

The capability of different methods to suppress complex noise, particularly the stripe artifacts common in push-broom sensors, is visualized via vertical profiles of the 36th band in Fig. 6. Our results are the most consistent with the ground truth in terms of pixel intensity while maintaining superior spatial continuity. Although certain baselines (e.g., "P+PD") perform adequately in homogeneous regions, they introduce significant artifacts at image boundaries or within high-frequency structures. In contrast, the proposed method achieves a superior balance between denoising and edge preservation, effectively protecting structural integrity under multisource degradation.

2) Visual Result Analysis: To intuitively demonstrate the restoration effects and performance differences of different methods, we present visualization reconstruction results for the PU, UR, and YT datasets in Figs. 7–9, respectively. Fig. 7 shows the restoration effects on the PU dataset. We can observe that, compared to methods using the HD framework, our method achieves better performance in spatial detail recovery. Although the four competing methods using the PD framework also perform well in spatial detail recovery, their restoration results exhibit obvious spectral distortion visually.

The results on the UR dataset shown in Fig. 8 more clearly confirm this view, where the overall color tones of the four PD-based methods show significant deviations from the true image in color restoration, highly consistent with the spectral infidelity conclusions revealed by the spectral curve plots in Fig. 6. Therefore, whether on the PU or UR dataset, our method achieves a better visual balance in spatial detail preservation and spectral fidelity, proving the effectiveness of IID constraints and frequency guidance mechanisms in suppressing spectral distortion. Fig. 9 shows the comparison results of different methods on the YT dataset. Since this dataset is an in situ measurement with clear original image details, we focus on evaluating overall restoration quality. As seen from the figure, under the extreme lunar environment, competing

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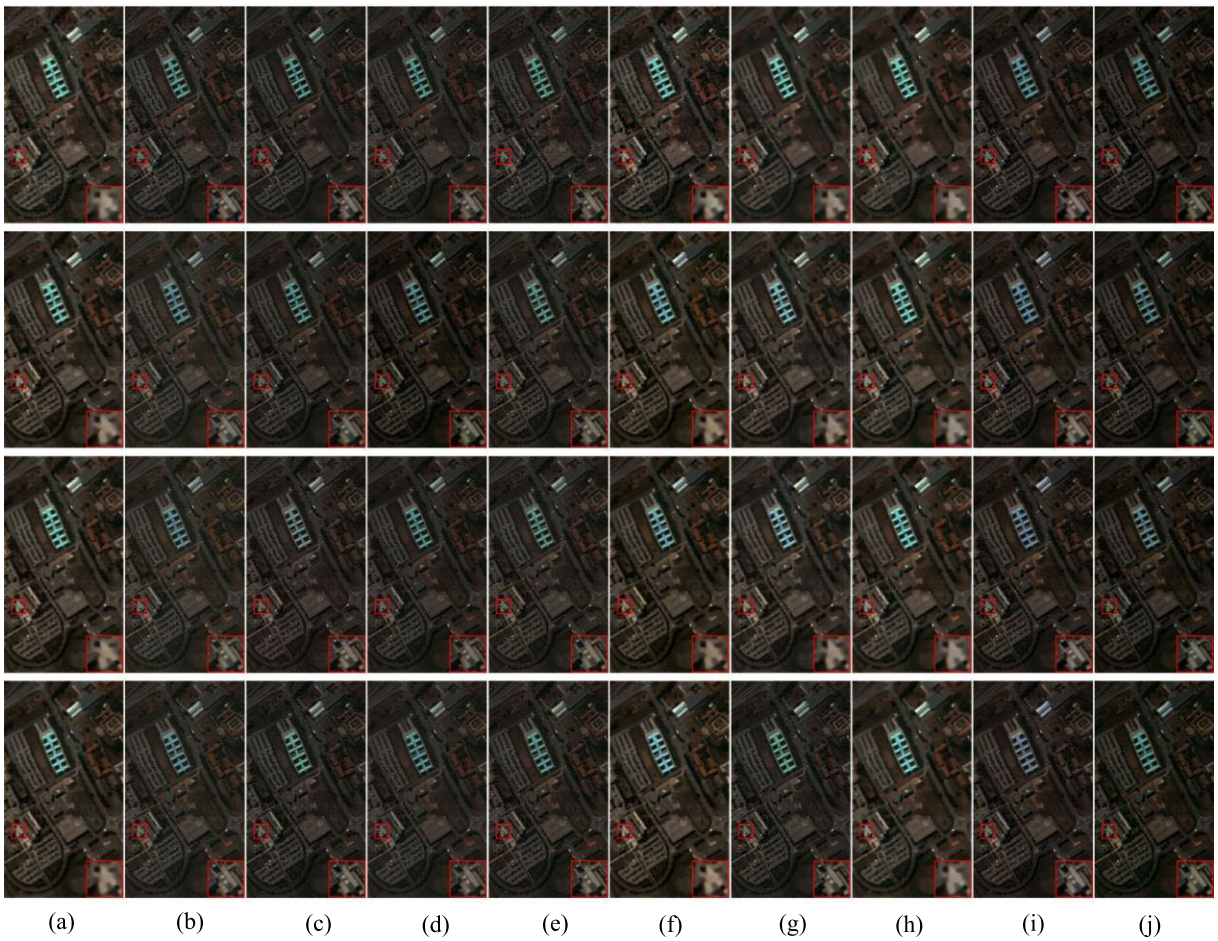


Fig. 7. Qualitative restoration results of different methods under four degradation cases on the PU dataset. Each column represents a specific method, while each row corresponds to a different degradation case: (a) A+PD, (b) B+PD, (c) N+PD, (d) P+PD, (e) A+PD, (f) B+HD, (g) N+HD, (h) P+HD, (i) UH, and (j) Ours.

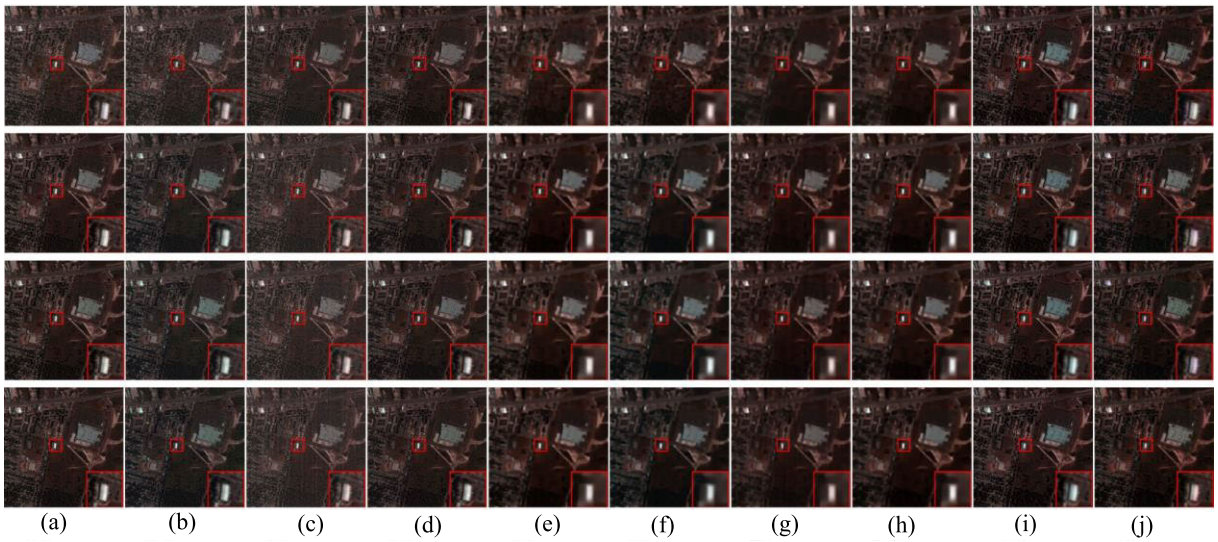


Fig. 8. Qualitative restoration results of different methods under four degradation cases on the UR dataset. Each column represents a specific method, while each row corresponds to a different degradation case: (a) A+PD, (b) B+PD, (c) N+PD, (d) P+PD, (e) A+PD, (f) B+HD, (g) N+HD, (h) P+PD, (i) UH, and (j) Ours.

methods generally exhibit poor robustness. HD-based methods exhibit extreme instability under lunar scenarios, with several experimental groups failing to produce effective restoration images. The remaining outputs suffer from severe blurring and

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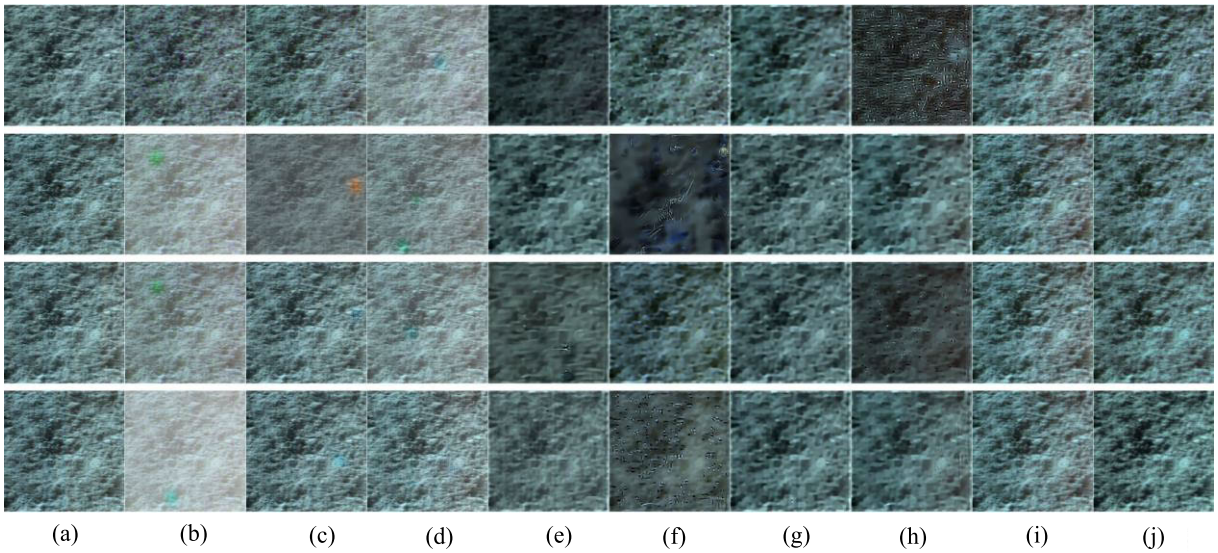


Fig. 9. Qualitative restoration results of different methods under four degradation cases on the YT dataset. Each column represents a specific method, while each row corresponds to a different degradation case: (a) A+PD, (b) B+PD, (c) N+PD, (d) P+PD, (e) A+HD, (f) B+HD, (g) N+PD, (h) P+HD, (i) UH, and (j) Ours.

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE PU DATASET

	Methods	A+PD	B+PD	N+PD	P+PD	A+HD	B+HD	N+HD	P+HD	UH	Ours
Case1	OA(%) ↑	74.34	71.10	81.66	79.91	75.54	74.04	82.93	83.44	**83.85**	83.45
	AA(%) ↑	61.54	57.31	70.89	66.45	60.92	59.18	72.64	73.22	**76.90**	75.16
	KAPPA(%) ↑	64.20	59.11	75.11	72.61	65.81	63.40	76.92	77.63	**78.42**	77.25
Case2	OA(%) ↑	81.32	77.65	81.10	82.79	83.80	83.81	83.72	85.65	86.15	**86.33**
	AA(%) ↑	72.04	67.03	72.31	76.75	74.68	73.76	74.37	77.87	**82.39**	82.00
	KAPPA(%) ↑	74.55	68.98	74.29	76.53	78.24	78.27	78.27	80.89	81.04	**81.32**
Case3	OA(%) ↑	79.74	77.57	81.09	83.17	86.16	83.79	83.36	85.49	85.58	**85.78**
	AA(%) ↑	71.57	66.82	72.44	77.14	80.39	73.60	73.95	77.85	**80.95**	80.38
	KAPPA(%) ↑	72.15	68.84	74.29	77.09	81.44	78.22	77.59	80.55	80.17	**80.58**
Case4	OA(%) ↑	79.22	76.47	79.53	79.61	82.31	82.75	83.93	82.41	85.28	**86.30**
	AA(%) ↑	69.68	63.87	69.90	72.34	72.32	71.08	73.70	74.44	**81.26**	81.05
	KAPPA(%) ↑	71.34	67.19	71.87	71.84	76.10	76.69	78.31	75.92	**81.87**	81.40

residual noise. This degradation is primarily attributed to the inherent lack of PAN-guidance in these diffusion-based priors. Consequently, the model's internal priors become misguided, leading to the collapse of spatial–spectral consistency and the production of physically inexplicable results. PD-based methods, with their low-rank decomposition assumptions, tend to oversmooth restoration results when facing sparse-featured lunar surface data, leading to severe loss of inherent texture information in the image. These issues may arise because lunar surface ground object features are sparse, causing the above low-rank-based methods to remove original high-frequency details and texture information of the image as noise while denoising high-frequency noise. This widespread performance degradation further highlights the advantages of the proposed IID-guided frequency decoupling method, which can effectively separate and preserve high-frequency spatial details, thereby better adapting to image restoration in extreme environments.

3) Application of Restoration Results to Downstream Classification and Unmixing Tasks: To verify the differences in ground object information fidelity and practical value of the restoration results, we conducted classification experiments on the PU dataset using support vector machine (SVM), and for the UR dataset, we performed unmixing using the FCLS method with given endmember spectra.

From the results in Tables IV and V, we observe that restoration fidelity metrics are not always linearly consistent with downstream task performance, a phenomenon also noted in joint restoration and classification frameworks [50]. For example, in the PU dataset under the Case 2 scenario, the "N+PD" baseline achieved competitive PSNR and SSIM, but underperformed other competing methods in classification accuracy. Similarly, in the UR dataset under Case 1, the SSIM of "P+HD" and "N+PD" methods differed by up to 20%, but their difference in the unmixing accuracy metric RMSE was less than 0.5%.

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TABLE V
SPECTRAL UNMIXING RESULTS OF DIFFERENT METHODS ON THE UR DATASET

	Methods	A+PD	B+PD	N+PD	P+PD	A+HD	B+HD	N+HD	P+HD	UH	Ours
Case1	RMSE ↓	0.2063	0.2361	0.1873	0.1904	0.1919	0.1898	0.1972	0.1898	0.1842	**0.1830**
	CC ↑	0.7175	0.6557	0.7735	0.7688	0.7711	0.7739	0.7646	0.7783	0.7832	**0.7884**
Case2	RMSE ↓	0.1820	0.1814	0.2117	0.1730	0.1803	0.1742	0.1922	0.1829	0.1725	**0.1724**
	CC ↑	0.7883	0.8023	0.7088	0.8127	0.8057	**0.8154**	0.7769	0.7984	0.8127	0.8143
Case3	RMSE ↓	0.1816	0.1823	0.2169	0.1791	0.1824	0.1805	0.1910	0.1871	0.1808	**0.1715**
	CC ↑	0.7891	0.8001	0.7068	0.8095	0.8002	0.8103	0.7722	0.7969	0.7927	**0.8163**
Case4	RMSE ↓	0.1841	0.1913	0.2138	0.1767	0.1821	0.1761	0.1987	0.1845	0.1900	**0.1754**
	CC ↑	0.7831	0.7792	0.7067	0.8041	0.8011	0.8007	0.7631	0.7942	0.7693	**0.8094**

TABLE VI
QUANTITATIVE RESULTS OF THE ABLATION STUDY ON THE CORE COMPONENTS OF THE PROPOSED METHOD

	PU				UR				YT			
	M1 ours	M2 w/o L_{IID}	M3 w/o L_P	M4 w/o AFMN	M1 ours	M2 w/o L_{IID}	M3 w/o L_P	M4 w/o AFMN	M1 ours	M2 w/o L_{IID}	M3 w/o L_P	M4 w/o AFMN
PSNR ↑	**30.75**	30.70	27.97	30.06	**29.49**	29.36	27.97	29.29	**33.89**	32.71	32.99	33.64
SSIM(%) ↑	84.79	**86.36**	77.37	83.90	84.37	**85.67**	77.37	84.25	92.08	90.75	90.01	**92.17**
ERGAS ↓	18.79	**18.49**	26.80	20.78	23.19	**23.13**	26.80	24.00	**7.39**	8.61	8.19	7.64
SAM ↓	**6.68**	7.14	8.73	6.91	**8.42**	9.47	8.73	8.92	**3.58**	4.18	3.68	3.76

These observations underscore the inherent conflict between spatial smoothness and spectral fidelity during ZSR. Conventional metrics like PSNR and SSIM are intensity-driven; they tend to favor models that prioritize spatial regularization and statistical averaging. However, such spatial smoothness often acts as a low-pass filter that inadvertently blurs narrowband spectral signatures. While this oversmoothing strategy effectively boosts perceptual quality scores, it leads to the loss of discriminative spectral features essential for pixelwise classification and endmember abundance estimation.

In contrast, our proposed method consistently yields superior performance across diverse downstream applications, achieving the optimal equilibrium between computational reconstruction quality and physical information authenticity. By explicitly navigating the inherent conflict between spatial smoothness and spectral fidelity, our IID-guided framework safeguards critical discriminative features that are often suppressed by purely statistical smoothing filters. This resilience ensures that the restored HSIs are not merely visually plausible, but remain high-fidelity data sources suitable for quantitative scientific analysis, such as precise mineral identification and subpixel abundance estimation in challenging extraterrestrial environments.

F. Ablation Studies

Ablation experiments are conducted to validate the effectiveness and necessity of each core component in our proposed framework. As discussed in the introduction, the synergy between IID-guided physical modeling and PAN-spectral fusion forms the backbone of our method.

Specifically, we evaluate the contributions of the AFMN for adaptive feature modulation, the intrinsic constraint losses ($\mathcal{L}_{\mathrm{R}}$ and $\mathcal{L}_{\mathrm{S}}$) for physical decoupling, and the pan-spectra decoupling fidelity loss ($\mathcal{L}_{\mathrm{P}}$) for spatial fidelity. To evaluate the contribution of each component, four configurations are compared: M1 represents the complete framework; M2 removes the intrinsic constraint losses $\mathcal{L}_{\mathrm{IID}}$ ($\mathcal{L}_{\mathrm{R}}+\mathcal{L}_{\mathrm{S}}$), which are treated as an indivisible entity due to their physically essential role in stable decoupling; M3 modifies the output of the CMSFM coefficient tensor to a single-channel format and computes its L_1 loss against the corresponding PAN image. This setup is designed to examine the necessity of CMSFM's multichannel adaptive coefficients. Accordingly, the input to this variant is changed to include only the HSI, and M4 removes the AFMN module.

TABLE VII
PERFORMANCE COMPARISON OF ABLATION VARIANTS ON THE DOWNSTREAM HYPERSPECTRAL UNMIXING TASK

Methods	M1	M2	M3	M4
RMSE ↓	0.1830	0.1981	0.1829	0.1833
CC ↑	0.7884	0.7570	0.7906	0.7877

The ablation results on Case 1 across three datasets are detailed in Table VI. A key observation, as noted in the comparison between M1 and M2, is that M2 yields comparable or occasionally slightly superior PSNR, SSIM, and ERGAS metrics, yet suffers from a severe degradation in SAM. This phenomenon reveals the inherent conflict between statistical intensity fitting and physical spectral fidelity. In the absence of IID constraints ($\mathcal{L}_{\mathrm{R}}$ and $\mathcal{L}_{\mathrm{S}}$), M2 possesses excessive degrees of freedom to minimize the pixelwise mean squared error. By performing aggressive spatial smoothing or fitting to stochastic noise residuals, M2 can mathematically boost PSNR. However, such optimization lacks a physical bottleneck, leading to nonphysical spectral shifts and a loss of the unique spectral profiles of ground objects. In contrast, M1 utilizes IID as a physical inductive bias, anchoring

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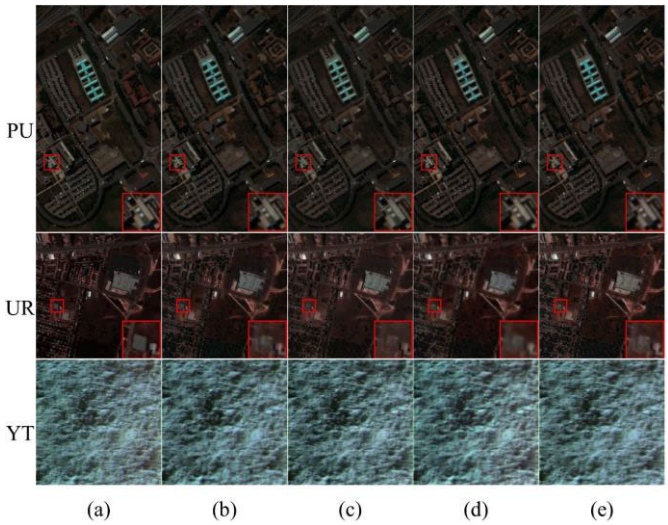


Fig. 10. Qualitative comparison of the ablation study results under Case 1 scenario. (a) Ground-truth image. (b)–(e) Restoration results of different ablation configurations, denoted as M1 through M4, respectively.

TABLE VIII
QUANTITATIVE COMPARISON OF DENOISING PERFORMANCE ON THE PU DATASET

	Methods	A	B	N	P	UH	Ours
Case1	PSNR↑	28.15	27.10	32.39	31.63	32.02	**32.71**
	SSIM(%)↑	85.36	82.29	94.84	92.63	94.18	**95.22**
	ERGAS↓	14.88	16.97	10.23	10.64	11.17	**10.94**
	SAM↓	5.59	6.39	3.65	3.59	3.42	**3.01**
Case2	PSNR↑	35.36	33.66	30.99	36.24	33.75	**36.80**
	SSIM(%)↑	94.28	94.15	90.53	**96.55**	96.20	96.25
	ERGAS↓	10.33	18.92	14.72	7.87	9.90	**6.62**
	SAM↓	4.42	6.11	7.51	2.49	2.64	**2.47**
Case3	PSNR↑	**35.54**	33.77	30.93	35.21	32.53	34.19
	SSIM(%)↑	95.17	94.38	90.54	**97.51**	95.11	95.72
	ERGAS↓	10.33	17.87	14.94	**6.87**	10.80	11.02
	SAM↓	4.09	5.98	7.54	**2.48**	3.11	2.53
Case4	PSNR↑	**35.12**	33.18	30.42	34.38	31.11	32.24
	SSIM(%)↑	95.42	94.17	90.54	**95.84**	93.48	94.94
	ERGAS↓	18.32	14.79	17.48	13.65	**12.37**	14.68
	SAM↓	5.13	5.73	7.25	**3.60**	3.80	4.05

the restoration within a reflectance–shading decomposition manifold. While this structural constraint may slightly limit the model's flexibility to fit every pixel intensity (leading to a marginal PSNR sacrifice), it significantly enhances the SAM metric. This ensures that the restored vectors maintain their directional integrity in the N-dimensional spectral space, which is far more critical for downstream quantitative analysis. Downstream unmixing results (see Table VII) show that M1 achieves lower RMSE and higher CC than M2. This validates that IID-based decoupling enhances spectral integrity, leading to more accurate abundance estimation compared to variants without physical constraints.

For M3, while the SAM remains stable, the spatial metrics exhibit consistent degradation. This confirms that the proposed $\mathcal{L}_{\mathrm{P}}$ fidelity loss and the corresponding CMSFM structure are vital for high-fidelity spatial reconstruction, particularly in bridging the spectral–spatial gap between PAN and HSI. While Table VII shows that M3 achieves slightly higher unmixing accuracy in specific metrics, M1 provides the best overall balance between spatial reconstruction and spectral fidelity. Finally, M4 shows a comprehensive decline across all four metrics. This demonstrates that the AFMN effectively enhances the HSI–PAN feature fusion accuracy through frequency-domain modulation, providing a universal performance boost to the entire restoration pipeline.

We present visualized restoration results in Fig. 10. The figure shows that the M2 method produces obvious color differences relative to the true image (e.g., rooftops in the UR dataset detail view), proving the critical role of intrinsic constraints in spectral fidelity. The M3 method appears overall coarser and blurrier, demonstrating the necessity of CMSFM for high spatial detail reconstruction. M4 shows improvement in some aspects compared to M2 and M3 but remains significantly weaker than the complete model M1 overall. The above visualization results are consistent with the quantitative data (see Table VI), jointly proving the unique roles of the four main modules proposed in this article. These modules collaborate with each other to achieve joint improvement in spatial and spectral fidelity in solving the HSI multidegradation task.

G. Discussion

1) Effectiveness of Denoising: Since both the proposed and the competing composite methods follow a sequential DN followed by SR pipeline, we report the quantitative denoising metrics on the PU dataset in Table VIII to isolate the impact of the initial stage. The results indicate that most unsupervised denoising methods achieve comparable performance, establishing a robust baseline for HSI noise suppression.

However, a critical observation emerges when comparing the intermediate denoising metrics with the final joint restoration results. For nonintegrated composite methods, a significant performance gap often exists between the two stages, underscoring the issue of error propagation. By treating denoising and SR as independent tasks, stochastic noise residuals from the first stage tend to accumulate and amplify during subsequent spatial enhancement.

In contrast, integrated frameworks generally exhibit superior procedural stability compared to modular baselines. Notably, our physics-driven model not only offers enhanced robustness over these modular strategies but also surpasses the end-to-end UH method in critical spectral and structural metrics. A representative case is observed in the highly degraded Case 4 scenario, where the UH method may initially yield a competitive SAM metric during isolated denoising; however, its performance significantly deteriorates in the final joint restoration, ultimately resulting in spectral fidelity inferior to our method. This discrepancy underscores that in environments with complex, intertwined multidegradations, purely statistical end-to-end paradigms often lack the necessary structural constraints to maintain consistency.

2) Sensitivity Analysis of the Pretraining Phase: To evaluate the necessity of the pretraining stage, we conducted a

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TABLE IX
PERFORMANCE COMPARISON UNDER DIFFERENT PRETRAINING ITERATIONS

	T_{pre}	0	200	400	600	800	1000
Case1	PSNR↑	28.91	29.47	29.44	**29.68**	29.49	29.07
	SSIM(%)↑	83.82	84.93	84.46	**85.34**	84.37	83.78
	ERGAS↓	24.82	23.66	23.75	**23.09**	23.19	24.08
	SAM↓	9.83	8.85	8.52	8.42	**8.42**	9.41
Case2	PSNR↑	28.68	29.53	29.67	**29.77**	29.69	29.55
	SSIM(%)↑	84.86	86.15	85.77	**86.26**	85.85	85.41
	ERGAS↓	24.13	23.46	23.12	**22.92**	22.97	23.14
	SAM↓	9.47	8.49	8.53	8.42	**8.40**	8.59
Case3	PSNR↑	28.10	29.67	29.65	**29.85**	29.76	29.55
	SSIM(%)↑	84.46	85.89	85.65	**86.30**	86.20	85.44
	ERGAS↓	23.93	23.06	**22.23**	23.03	22.99	23.17
	SAM↓	9.31	8.61	8.50	8.44	**8.40**	8.62
Case4	PSNR↑	28.34	29.30	29.28	**29.49**	29.17	29.37
	SSIM(%)↑	84.38	85.32	85.04	**85.73**	84.62	85.19
	ERGAS↓	25.83	24.02	24.13	**23.47**	24.63	23.65
	SAM↓	9.99	9.05	8.80	8.88	**8.74**	8.83

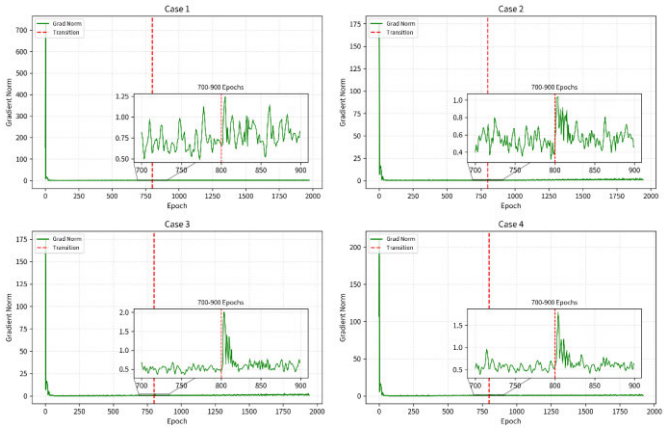


Fig. 11. Training stability analysis of the proposed framework.

comprehensive sensitivity analysis on the UR dataset by varying the number of pretraining iterations (T_{pre}). The quantitative results across four degradation scenarios are summarized in Table IX.

As shown in the results, omitting the pretraining stage ($T_{pre} = 0$) consistently leads to suboptimal performance, confirming that the low-resolution pretraining provides a superior initialization that effectively avoids poor local minima. Notably, even without pretraining, our framework still outperforms state-of-the-art methods, indicating the inherent robustness of the proposed architecture. The restoration performance typically peaks at $T_{pre} \in [600, 800]$ and shows a marginal decline thereafter, suggesting diminishing performance gains from excessive pretraining. Considering all degradation scenarios, $T_{pre} = 800$ consistently provides the best or near-best SAM performance and is therefore adopted in our final implementation.

Furthermore, to verify the stability of the proposed two-stage optimization, Fig. 11 presents the evolution of the gradient norm around the transition point from the pretraining stage to the joint optimization stage. It can be observed that the gradient norm remains stable during pretraining and exhibits only a modest, controlled increase immediately following the transition, followed by rapid convergence to a steady state. No abrupt gradient explosion or optimization oscillation is observed under any noise setting. These observations indicate that the pretrained denoising network provides an effective initialization, facilitating a smooth and stable transition to the joint optimization phase while maintaining robust convergence across varying noise conditions.

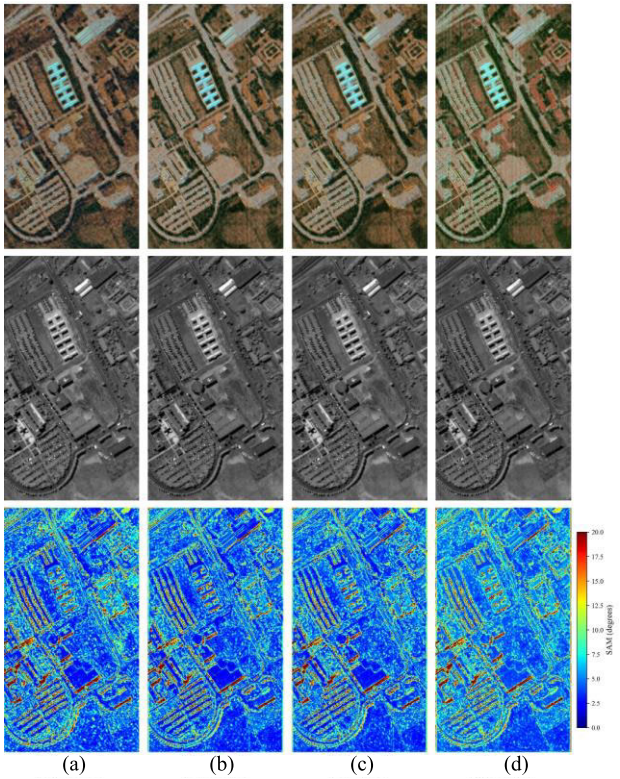


Fig. 12. Visualization of intrinsic decomposition components and pixelwise SAM error distributions: (a) Case 1, (b) Case 2, (c) Case 3, and (d) Case 4.

3) IID Components and SAM Error Visualization: Fig. 12 illustrates the IID decomposition results and the corresponding SAM error maps under various degradation scenarios. As shown, the reflectance component effectively discriminates between different land-cover types with high spectral purity, while the shading component maintains a stable topographic structure aligned with the PAN imagery.

The SAM error maps provide further insight into the physical restoration performance. Across all cases, our method achieves superior spectral recovery within homogeneous objects, with SAM values consistently maintained below 5°. Although localized errors appear at object boundaries due to abrupt spectral transitions, these errors are effectively contained and do not propagate into the global reconstruction. Notably, in Case 4, the visualization suggests that a considerable portion of the degradation is captured by the shading component; this prepurification allows the reflectance stream to avoid spectral collapse. These visualizations are consistent with the intended physical decomposition mechanism of the proposed framework.

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TABLE X
COMPARISON OF INFERENCE TIME FOR DIFFERENT METHODS

Method	A+PD	B+PD	N+PD	P+PD	A+HD	B+HD	N+HD	P+HD
Time(s)	258.87	309.21	293.39	249.49	34.27	84.61	68.79	24.89
Method	A	B	N	P	HD	PD	UH	Ours
Time(s)	11.87	62.21	46.39	2.49	247.00	22.40	326.92	1100.20

4) Computational Efficiency: Table X presents a comparative analysis of the inference time for different methods on the PU dataset. Among the evaluated frameworks, the proposed method requires the longest execution time. This increased computational overhead is primarily attributed to the multiobjective optimization involved in maintaining several intrinsic constraint losses, coupled with a larger number of iterations epochs required to ensure stable physical decoupling. Our method provides a robust solution that prioritizes restoration quality over computational speed, which is a justifiable tradeoff in offline data analysis scenarios where physical consistency is paramount.

V. CONCLUSION

This article investigates the problem of HSI restoration under coupled multidegradation conditions, with a particular focus on the high-fidelity demands of deep-space exploration. By explicitly modeling the physical imaging mechanism, we propose an unsupervised ZSR framework based on IID, which reformulates multidegraded HSI restoration as a reflectance–shading decoupling problem. This physics-driven formulation enables spatial enhancement while preserving physically meaningful spectral reflectance. To ensure stable intrinsic decomposition without external supervision, a progressive optimization framework is constructed by jointly introducing intrinsic physical consistency constraints, frequency-guided feature modulation, and PAN-guided spatial fidelity modeling. These components collaboratively enforce the physical assumptions of IID, allowing effective separation of spectral properties and spatial variations under complex degradation scenarios. Extensive experiments on public remote sensing benchmarks and real lunar in situ HSI demonstrate that the proposed method consistently outperforms existing approaches in terms of spectral fidelity, spatial detail preservation, and robustness, effectively serving as a high-precision spectral ruler for complex and nonterrestrial imaging conditions. Ablation studies further verify the necessity and complementarity of each modeling component. Overall, this work provides a physically interpretable and practically effective zero-shot paradigm that aligns with the strategic requirements for high-fidelity data processing in future deep-space missions.

Future work will focus on two directions. First, since the current framework is developed based on near-Lambertian surface assumptions and stable observation geometries, we aim to integrate explicit radiative transfer models (e.g., the Hapke model) into the DIP-based framework to further enhance its physical generalization on non-Lambertian surfaces with complex viewing geometries. Second, we will explore extending the framework to more challenging scenarios, such as multidegradation restoration from a single hyperspectral observation without auxiliary PAN guidance.

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